



**Verified Carbon
Standard**

IMPROVED COOKING STOVE PROGRAMME IN BURUNDI

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#CashForClimate

Document Prepared by AERA Group on behalf of Observatoire
Burundais de l'Environnement et de la Nature (OBEN) and POWA S.A.

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The purpose of this grouped project is to combat climate change and contribute to sustainable development of Burundi, by reducing non-renewable wood fuel consumption and greenhouse gas (GHG) emissions of household users in rural and/or peri-urban areas of Burundi by selling affordable Improved Cooking Stoves (ICSs) in replacement of traditional wood stoves. Thanks to the higher efficiency of ICS distributed compared to baseline wood stoves, the project will reduce the consumption and combustion of wood fuel more efficiently, thus reducing CO₂ emissions regarding the baseline situation. Although the project aims to replace baseline inefficient wood stoves by ICS, for emission reduction calculation, as per para 23 of the methodology AMS II-G v10.0 it is assumed that in the absence of the project activity, the baseline scenario would be the use of fossil fuels to meet similar thermal energy needs as those provided by the project devices.

The PoA and the CPA implementer is POWA S.A since September 2021¹, the main technology supplier under the PoA, with the international assistance of AERA Group, leading carbon finance advisor in Africa. POWA S.A will run the project in compliance with the PoA requirements, disseminating estimated 500,000 affordable Improved Cooking Stoves (ICSs) in replacement of traditional wood fuel cooking stoves, such as 3-stone fires, resulting in an average of 506,373 tCO₂ avoided per year or 3,544,611 tCO₂ over the 7-year crediting period.

Initially undertaken as CDM PoA 10474 “Improved Cooking Stove Programme in Burundi supported by Republic of Korea”, ECOEYE Co., Ltd. and/or Korean Party(ies) provide all implementation cost for all the stoves disseminated in the PoA until September 2021 (stove subsidy to make ICS affordable to household) and also provides all operation & maintenance cost of ICS manufacturing and monitoring for OBEN CPA implementer to operate the project under financially sustainable condition). Since September 2021, the, alternate project implementer POWA S.A is tasked with continuous cookstoves dissemination and project implementation.

The first instance to be included in the grouped project corresponds to the CPA 1 of the CDM PoA 10474 “Improved Cooking Stove Programme in Burundi supported by Republic of Korea”, for its post-2020 accounting of Emission Reductions.

This CPA qualifies as a micro-scale project type II which CDM units aims to achieve energy savings at a scale of no more than 600 MWh per year, which is equivalent to 1,800 MWh_{th} of annual energy savings per appliance.

To date 202,856 cookstoves have been distributed reducing 632,878 tCO₂e.

¹ Before this date, OBEN E.A.C was the project implémenter of the CPA.

1.2 Sectoral Scope and Project Type

Sectoral Scope: 03 Energy demand

Project Type: Type II – Energy Efficiency Improvement Project

Methodology used: AMS-II.G Ver.10.0- Energy efficiency measures in thermal applications of non-renewable biomass/

1.3 Project Eligibility

As per para. 2.1.1 of the VCS Standard, V.4.2, the project is eligible under the scope of the VCS Program because the VCS Scope includes project activities involving the following elements:

Elements	Project applicability
The six Kyoto Protocol greenhouse gases	The project activity and AMS II.G V.10.0 aims at CO ₂ reduction of non-renewable biomass.
Ozone-depleting substances.	n/a
Project activities supported by a methodology approved under the VCS Program through the methodology approval process.	n/a
Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval.	CDM methodology AMS II.G, V.10 is applied. The CDM is a VCS approved GHG program. ² Any methodology developed under the <u>United Nations Clean Development Mechanism</u> can be used for projects and programs registering with VCS. ³
Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements.	n/a

Furthermore, it cannot reasonably be assumed that the programme generates GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction. All users involved in the programme activity are fully independent from the project proponent.

Finally, the programme is taking place in Burundi, which is a LDC, the programme of activity is not excluded by the VCS Program under the circumstances indicated in Table 1 of para. 2.1.1 of the VCS Standard, V.4.2

1.4 Project Design

² <https://verra.org/project/vcs-program/projects-and-jnr-programs/develop-a-project/>

³ <https://verra.org/methodologies/>.

Please also refer to VCS Program Guide V.4.0, section 7.

The first instance⁴ of the grouped project disseminate improved cook stoves over time amongst households in Burundi.

Eligibility Criteria

For the inclusion of new project activity instances i.e., ICS, the project proponent shall ensure that it meets the eligibility criteria below.

No.	Criterion	How the new project activity instances to comply
1	Meet the applicability conditions set out in the methodology applied to the project	New project activity instances will meet the applicability conditions set out in Section 3.2 where the target of the end-user is household and the ICS deployed is at least 20% of thermal efficiency.
2	Use the technologies or measures specified in the project description.	Only improved cookstoves to be adopted in the project,
3	Apply the technologies or measures in the same manner as specified in the project description.	Only improved cookstoves to be adopted in the project and it will replace traditional cookstoves in household
4	Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	The new project activity instances will be installed within Burundi only and subject to the same baseline scenario determined in Section 3.4.
5	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.	All the new project activity instances that will be included in the grouped project will respect additionality criteria set out in section 3.5.

1.5 Project Proponent

Organization name

POWA S.A

⁴ The first instance to be included in the grouped project corresponds to the CPA 1 of the CDM PoA 10474 “Improved Cooking Stove Programme in Burundi supported by Republic of Korea”, for its post-2020 accounting of Emission Reductions

Role in the project	Project implementer as from September 2021
Contact person	Pascal Rwemera
Title	Director
Address	Mumiruwa 105, Kinindo, Muha, Bujumbura-Mairie
Telephone	+257-76-660-334
Email	rwemerapascal@gmail.com

1.6 Other Entities Involved in the Project

Organization name	Observatoire de l'Environnement et de la Nature « OBEN »
Contact person	Claver Ndizeye
Title	Director
Address	Avenue Nyabisindu, Q9 Bujumbura
Telephone	+257-76-602-600
Email	loicqueen@yahoo.fr

Organization name	AERA Group SAS
Role in the project	Carbon consultant and trader
Contact person	Alexandre Dunod
Title	COO
Address	28 cours Albert 1er, Paris, 75008, France
Telephone	+33642960978
Email	a.dunod@aera-group.fr

1.7 Ownership

As per VCS Program Definitions version 4, the project ownership is the legal right to control and operate the project activities. As project implementer, POWA S.A have the full ownership of the project. Furthermore, as per VCS Standard V4.2 para 3.6.1, the project description shall be accompanied by one or more of the following types of evidence establishing project ownership accorded to the project proponent(s), or program ownership accorded to the jurisdictional proponent(s), as the case may be (see the VCS Program document Program Definitions for definitions of project ownership and program ownership). The type of evidence to prove project ownership is point 5 of para 3.6.1 :“An enforceable and irrevocable agreement with the holder of the statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals which vests project ownership in the project proponent”.

All the end-users (technology owners), sign a stove sales agreement, which contains user information, rights and commitments including relinquishing ERs ownership to POWA S.A.

Also, OBEN and POWA has entered in agreement with AERA Group, in which they cede rights to sell ERs generated by their programme. End-users' agreements and agreements extract between OBEN and AERA as well as POWA and AERA are available and will be provided to DOE to prove project ownership.

To note that Ecoeye (registered formerly as CME under CDM before conversion) is no longer involved in the VCS conversion. Ecoeye financed the project under CDM, to convert the credits on the Korean market (KOC). Due to the end of the CDM, and the project being converted to VCS, from 4th verification and for the rest of the program Ecoeye is no longer involved and it is POWA SA in Burundi that continues the distribution and takes the role under the voluntary scheme. AERA Group has an agreement to retrocede revenues to ECOEYE for the part of the credits generated by the stoves they financed

1.8 Project Start Date

02-November-2018 (the date of the first ICS emission reductions).

1.9 Project Crediting Period

This project activity is registered under CDM mechanism⁵ with a renewable crediting period of 07 years x 3, total duration of 21 years. Its VCS Crediting Period shall be likewise, seven (7) years renewable 2 times for a total of 21 years maximum.

First crediting period starts from 02-November-2018 to 01-November-2025.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

5

https://cdm.unfccc.int/ProgrammeOfActivities/poa_db/U2ZYTf1EWXPBHK0069GCLNSDRIQ78A/view

Project Scale	
Project	
Large project	x

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
2018	0
2019	165,159
2020	358,566
2021	832,714
2022	897,593
2023	684,862
2024	402,457
2025	203,260
Total estimated ERs	3,544,611
Total number of crediting years	7
Average annual ERs	506,373

1.11 Description of the Project Activity

n/a. Please see registered CDM-CPA (as per para. 3.20.5 of VCS Project Standard V.4.2).

Link to the CPA:

https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R5MF4X6GWEJY8CVS9317ADKB2NUHTP/view

1.12 Project Location

The geographical boundaries of the CPA comprise rural and/or peri-urban areas of the whole Republic of Burundi (18 administrative provinces: geographic coordinates 3° 23' S, 29° 55' E).



Figure 1: Map of Burundi

Rural population represents 89,9% of Burundi total population as per table 5 page 7 of Central Census Bureau analysis conducted in 2008⁶.

1.13 Conditions Prior to Project Initiation

The conditions prior to the project initiation are the same as the baseline scenario.

The project activity involves dissemination of higher efficiency biomass fired cook stoves use as household appliances for cooking purpose. This will contribute to the reduction of non-renewable biomass consumption which would have been otherwise consumed by traditional three stone open firing, less efficient cook stoves.

Please refer to section B.3 of CDM-CPA-DD of the UNFCCC registered CDM PoA Ref No. 10474.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

There are no laws and regulations governing the use of ICSs in Burundi. The project is a voluntary effort by POWA S.A, OBEN, AERA and Ecoeye.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

⁶https://www.usaid.gov/sites/default/files/documents/1866/2008%20Burundi%20Population%20Census_Summary%20of%20Final%20Results.pdf

The project activity has already been registered under Clean Development Mechanism (CDM) of UNFCCC, under Kyoto Protocol. This project is a Component Project Activity (CPA) under a registered Programme of Activities (CDM-PoA). The details are as follows:

CDM-PoA Registration number: 10474.

Title: Improved Cooking Stove Programme in Burundi supported by Republic of Korea

Weblink:

https://cdm.unfccc.int/ProgrammeOfActivities/poa_db/U2ZYTf1EWXPBHK0069GCLNSDRIQ78A/view

CPA Reference number: 10474-P1-0001- CP1

Title: Improved Cooking Stoves Programme in Burundi supported by Republic of Korea – CPA1

Weblink:

https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R5MF4X6GWEJY8CVS9317ADKB2NUHTP/view

Previous issuance requests:

- Monitoring report: 10 Sep 2019 - 15 Sep 2019 | CERs requested : 826
- Monitoring report: 16 Sep 2019 - 30 Apr 2020 | CERs requested : 81 469
- Monitoring report: 01 May 2020 - 31 Dec 2020 | CERs requested : 183 191.

1.15.2 Projects Rejected by Other GHG Programs

The Project is not rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Burundi is non-annex 1 country and there is no compliance with an emission trading program or to meet binding limits on GHG emissions for this project activity. The project activity has been registered as a CPA under a CDM-PoA (Ref. 10474) and GHG emission reductions and removals generated by the project will not be used for any compliance under any programs or mechanisms, except CDM.

The project activity under VCS shall not claim emission reductions for the same period that has been/will be covered under CDM as a part of the registered PoA. Thus, there will not be any possibility of double accounting of emission reductions.

1.16.2 Other Forms of Environmental Credit

The project has neither sought nor received another form of GHG-related environmental credits.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

The country's Poverty Reduction Strategy Paper II (PRSP II) for 2012–2016, extended to 2018 and anchored in Burundi Vision 2025, includes 'promoting development through sustainable environmental and space management' as its 4th pillar, which chief component of (ii) protection of the environment and sustainable resource management.

As also supported by the World Food Program's Burundi interim country strategic plan (2018– 2020), fuel-efficient stoves are promoted for helping to reduce respiratory diseases, the environmental impact of deforestation and the protection risks associated with collecting fuelwood.

Table 1: Impact on Sustainable Development Goals in Burundi

SDG	Program Alignment
3: Ensure healthy lives and promote well-being for all at all ages	contributes to SDG 3, as it reduces the number of deaths and illnesses from air pollution. The ICSs help cut carbon monoxide and other particles related to incomplete and/or indoor solid fuels combustion and control the fire better reducing accidents. Smoke from combusted wood fuel is the leading cause of acute respiratory infections and eye problems (Peprah, 2010). Other illnesses are lung cancer and asthma, which concern especially women and children, causing more than 1.5 million deaths a year (Rob Bailis, 2009) (World Health Organization, 2006). This SDG is monitored by the percentage of people experiencing less smoke thank to the ICS introduction
7: Ensure access to affordable, reliable, sustainable and modern energy for all	contributes to the goal of doubling the global rate of improvement in energy efficiency by 2030. The project aims to replace inefficient cooking stoves by more efficient cooking stoves (ICSs), an upgraded technology. This SDG is monitored by the number of households equipped with ICS
8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	creates more and better employment (production, marketing, distribution, after-sale services) throughout the PoA lifecycle. Furthermore, ICSs promote safe and secure working environments of kitchen workers. ICSs increase economic productivity through improved technology. Most important benefits are reduced workload in wood fuel procurement and/or reduced expenses for wood fuel purchase.

	This SDG is monitored by the number of decent work created.
13: Take urgent action to combat climate change and its implications.	Apart from CO2 emission reductions, strengthens resilience and adaptive capacity to climate-related hazards and natural disasters due to the increase of financial savings of households or communities.

1.17.2 Sustainable Development Contributions Activity Monitoring

During this monitoring period 55,612 households have been equipped with ICS for a total of 202,856 household equipped since the project start, resulting in 632,878 tCO2e avoided.

Monitoring surveys conducted during this monitoring period also highlighted that 98% of the end-users surveyed experienced less smoke and less pollution thanks to the introduction of the ICS, thus reducing illness from indoor air pollution the follow.

Table 2 : Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
<i>Sequential row number</i>	<i>SDG Target number</i>	<i>Number and text of SDG indicator or, if no official SDG indicator is applicable, user-defined indicator</i>	<i>Indicate the project's contribution to the SDG Indicator (implemented activities to increase or decrease)</i>	<i>Brief description of the quantifiable impact of the project's activities related to the SDG indicator, during the monitoring period.</i>	<i>Brief description of the cumulative quantifiable impact of the project's activities related to the SDG indicator, over the project lifetime.</i>
1)	3	Proportion of users experiencing less smoke and time saving while using the cookstives	Implemented activities to increase	98% of the end-users surveyed experience less fumes when cooking with the ICS distributed	98% Of the end-users experience less fumes during the overall project
2)	7	Number of Households equipped with ICS	Implemented activities to increase	55,612 Households equipped with ICS	202,856 Households equipped with ICS during the project lifetime
3)	8	Number of decent Jobs created	Implemented activities to increase	No new jobs were created during this monitoring period	The project is expected to create 30 Jobs during his lifetime.
4)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	By conserving distributing 202,856 cookstoves, the programme avoided the emissions of 632,878 tCO ₂ e during this monitoring period	The expected emissions reductions thanks to the project is 3,544,611 during the project lifetime.

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable to this project activity.

The project activity has been designed to include only new improved cookstoves. No used improved cookstoves will be transferred or deployed from outside the geographical project boundary to the project activity. Also, the project activity applies CDM Methodology: AMS-II.G. "Energy Efficiency Measures in Thermal Applications of Non-Renewable Biomass", Version 10.0 According to the para 34 of the methodology, the default net to gross adjustment factor of 0.95 has been applied to account for leakage and therefore no surveys or no separate leakage management are required. This information has been further referred under the registered CPA-DD, under the section B.4.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not Applicable.

2 SAFEGUARDS

2.1 No Net Harm

This section is not applicable as project activity is a CDM registered GHG program, under CDM-PoA 10474. Thus, section 3.20.5 of the VCS Standard V4.2 has been referred.

2.2 Local Stakeholder Consultation

This section is not applicable as project activity is a CDM registered GHG program, under CDM-PoA 10474. Thus, section 3.20.5 of the VCS Standard V4.2 has been referred.

Please refer to sections F of PoA-DD of the UNFCCC registered CDM project with reference no. 10474.

2.3 Environmental Impact

This section is not applicable as project activity is a CDM registered GHG program, under CDM-PoA 10474. Thus, section 3.20.5 of the VCS Standard V4.2 has been referred.

Please refer to sections E of PoA-DD of the UNFCCC registered CDM project with reference no. 10474.

2.4 Public Comments

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The project was opened for public comments spanning from 01/04/2022 to 01/05/2022. However, no comments have been received during the public comment period as verifiable on project webpage at “<https://registry.terra.org/app/projectDetail/VCS/2828>”

2.5 AFOLU-Specific Safeguards

The project is a non-AFOLU project activity, hence not applicable.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

This section is not applicable as project activity is a CDM registered GHG program, under CDM-PoA 10474. Thus, section 3.20.5 of the VCS Standard V4.2 has been referred.

Information related to Methodology application can be referred from the Section B of the registered PoA-DD.

3.2 Applicability of Methodology

This section is not applicable as project activity is a CDM registered GHG program, under CDM-PoA 10474. Thus, section 3.20.5 of the VCS Standard V4.2 has been referred.

Information related to Methodology application can be referred from the Section I of the registered PoA-DD.

3.3 Project Boundary

This section is not applicable as project activity is a CDM registered GHG program, under CDM-PoA 10474. Thus, section 3.20.5 of the VCS Standard V4.2 has been referred.

Information related to Methodology application can be referred from the Section B of the registered PoA-DD.

3.4 Baseline Scenario

This section is not applicable as project activity is a CDM registered GHG program, under CDM-PoA 10474. Thus, section 3.20.5 of the VCS Standard V4.2 has been referred.

Information related to Methodology application can be referred from the Section B of the registered PoA-DD.

3.5 Additionality

This section is not applicable as project activity is a CDM registered GHG program, under CDM-PoA 10474. Thus, section 3.20.5 of the VCS Standard V4.2 has been referred.

Information related to Methodology application can be referred from the Section C of the registered PoA-DD.

3.6 Methodology Deviations

A temporal deviation has been applied during this monitoring period, regarding the monitoring of the efficiency that should be monitored on an annual frequency. PP was not able to conduct efficiency test on an annual basis, because of some delays observed with the laboratory.

To ensure that this deviation does not negatively impact the conservativeness of the quantification of GHG emission reductions, it was decided for this monitoring period to use the lowest efficiency of all the batches as the efficiency for all the batches. This results in a reduction of 6,336 tCO₂e

Also, this temporary deviation relates only to the procedures for monitoring the efficiency, and does not relate to any other part of the methodology.

4 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

This section is not applicable as project activity is a CDM registered GHG program, under CDM-PoA 10474. Thus, section 3.20.5 of the VCS Standard V4.2 has been referred.

Information related to Methodology application can be referred from the Section B.4 of the CPA-DD

4.2 Project Emissions

As per AMS II.G v10.0 methodology, Emissions reductions are calculated directly. Therefore, this section is not applicable

4.3 Leakage

According to the para 34 of the methodology, the default net to gross adjustment factor of 0.95 has been applied to account for leakages. No others leakage is considered. Therefore, the section is not applicable.

4.4 Estimated Net GHG Emission Reductions and Removals

This section is not applicable as project activity is a CDM registered GHG program, under CDM-PoA 10474. Thus, section 3.20.5 of the VCS Standard V4.2 has been referred.

Information related to net emission reductions can be referred from the Section B.4.4 of the registered CPA-DD

5 MONITORING

5.1 Data and Parameters Available at Validation

This section is not applicable as project activity is a CDM registered GHG program, under CDM-PoA 10474. Thus, section 3.20.5 of the VCS Standard V4.2 has been referred.

Information related to Methodology application can be referred from the Section B.4.2 of the CPA-DD

5.2 Data and Parameters Monitored

This section is not applicable as project activity is a CDM registered GHG program, under CDM-PoA 10474. Thus, section 3.20.5 of the VCS Standard V4.2 has been referred.

Information related to Methodology application can be referred from the Section B.5.1 of the CPA-DD

5.3 Monitoring Plan

This section is not applicable as project activity is a CDM registered GHG program, under CDM-PoA 10474. Thus, section 3.20.5 of the VCS Standard V4.2 has been referred.

Information related to Methodology application can be referred from the Section B.5.1 of the CPA-DD

6 ACHIEVED GHG EMISSION REDUCTIONS AND REMOVALS

6.1 Data and Parameters Monitored

Data / Parameter	$N_{y,j}$
Data unit	units
Description	Number of project devices of type i and batch j operating during year y

Value applied:					
	Year	Batch 1	Batch 2	Batch 3	Batch 4
	2021	42,495	88,443	56,676	10,873
	2022	42,495	88,443	56,676	10,873
Comments	Ny_{i,j} for each batch, is determined by multiplying all devices sold in the batch with the proportion of cooking stoves found to be operating in a representative sample of each batch (Pop_{stoves,y}). Batch 1 covers stoves sold between 02/11/2018 and 01/11/2019; Batch 2 covers stoves sold between 02/11/2019 and 01/11/2020; Batch 3 covers stoves sold between 02/11/2020 and 01/11/2021, and Batch 4, stoves sold between 02/11/2021 and 01/11/2022. However, as cut-off date is 31/03/2022, batch 4 represents stoves sold between 02/11/2021 and 01/11/2022. The monitored proportion of stoves found operational for each batch are represented in the table below:				
	Year	Batch 1	Batch 2	Batch 3	Batch 4
	2020	96,77%	98,33%	97,50%	100%
	2021	96,77%	98,33%	97,50%	100%

Data / Parameter	μ_y				
Data unit	Fraction				
Description	Adjustment to account for any continued use of pre-project devices during year y				
Value applied:	Year	Batch 1	Batch 2	Batch 3	Batch 4
	2021	94,67%	97,79%	99,21%	99,64%
	2022	94,67%	97,79%	99,21%	99,64%
Comments	During the annual monitoring campaign, mandated field agents inquire if the baseline stove that was supposed to be replaced by the ICS is still used. Field agents estimate the usage rate of the pre-project stove(s) by formulating questions to determine the frequency of usage of both the project devices and baseline devices				
	Sampling have been conducted following a 95/10 precision (Biennial inspection) in accordance with the “Standard for sampling and surveys				

	for CDM project activities and programme of activities”, and 95/10 precision was met.
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Data / Parameter	$\eta_{new,i,j}$				
Data unit	%				
Description	Efficiency of the device of each type i and batch j implemented as part of the project activity.				
Value applied:	Year	Batch 1	Batch 2	Batch 3	Batch 4
	WBT results 2021	30.71%	31.48%	31.61%	31.59%
	WBT results 2022	30.71%	31.48%	31.61%	31.59%
	Conservative Value chosen (2021)	30.71%	30.71%	30.71%	30.71%
	Conservative Value chosen (2022)	30.71%	30.71%	30.71%	30.71%
Comments	Tests are performed by third party “Centre de Recherche Universitaire sur les Energies Alternatives (CRUEA)” (University of Burundi) with calibrated equipment and cross-checked with manufacturer information suggesting an efficiency of 32.10%. Although this parameter should be monitored annually, PP was not able to conduct an annual monitoring. Therefore, sampling test have been conducted following a 95/10 precision (Biennial inspection) in accordance with the “Standard for sampling and surveys for CDM project activities and programme of activities”, and 95/10 precision was met. As PP was not able to ensure compliance with the annual monitoring of the efficiency, and to avoid an overestimation of emission reduction, PP decided to use the lowest value of all the WBTs results as efficiency value for all the batches. The value of 30.71% was therefore applied to all the batches.				

Data / Parameter	<i>Date of commissioning of batch j</i>
Data unit	Date
Description	To establish the date of commissioning, the Project Participant opts to group the devices in “batches” and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch
Value applied:	Excel spreadsheet provided to the VVB
Comments	

Data / Parameter	Date of commissioning of project device <i>i</i>				
Data unit	Date				
Description	Actual date of commissioning of the project device.				
Value applied:	Batch	Batch 1	Batch 2	Batch 3	Batch 4
	Date	02/11/2018	02/11/2019	02/11/2020	02/11/2021
Comments					

Data / Parameter	N				
Data unit	Number				
Description	Number of project devices distributed				
Value applied:	202,856 cookstoves full use equivalent				
	Batch	Batch 1	Batch 2	Batch 3	Batch 4
	N	43,912	89,942	58,129	10,873
Comments	Every time an ICS is sold a sale agreement is filled and an electronic database is filled. Based on the information collected				

	into this electronic database, the number of ICSs distributed is determined.
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Data / Parameter	NCV Biomass
Data unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass used in project devices
Value applied:	0.0156
Comments	Data from 2006 IPCC Guidelines for National Greenhouse Gas Inventories.

Data / Parameter	Life Span
Data unit	Number of years
Description	The operating lifetime of the project device. The life span should be reported in cases where the CPA implementer opts to account the efficiency loss as per paragraph 32 (a) of AMS-II.G.
Value applied:	N/A
Comments	Only reported if option a) of para. 32 of AMS-II.G. is chosen

Data / Parameter	$N_{d,HH}$
Data unit	Number
Description	Number of project devices distributed per household
Value applied:	1
Comments	Only one cooking stove per household is registered in the electronic database. If a household purchases more than one cooking stoves, monitoring surveys of sampled kitchens' stoves

in use will account for any additional project device and be reflected in adjustment factor $N_{d,HH}$.

6.2 Baseline Emissions

Emission reductions are calculated as:

$$ER_y = \sum_i \sum_j ER_{y,i,j} - LE_y$$

Where:

- i = Indices for the situation where more than one type of project device is introduced to replace the pre-project devices⁷
- j = Indices for the situation where there is more than one batch of project device
- ER_y = Emission reductions during year y in t CO₂e
- $ER_{y,i,j}$ = Emission reductions by project device of type i and batch j during year y in t CO₂e
- LE_y = Leakage emissions in the year y

$$ER_{y,i,j} = B_{y,savings,i,j} \times N_{y,i,j} \times \mu_y \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected_fossil\ fuel}$$

Where:

- $B_{y,savings,i,j}$ Quantity of woody biomass that is saved in tonnes per cook stove of type i and batch j during year y
- $f_{NRB,y}$ Fraction of woody biomass saved by the project activity in year y (established as non-renewable biomass using survey methods)
- $NCV_{biomass}$ Net calorific value of the non-renewable woody biomass that is substituted (IPCC default for wood fuel, 0.015 TJ/tonne, based on the gross weight of the wood that is 'air-dried')
- $EF_{projected_fossilfuel}$ Emission factor for the fossil fuels projected to be used for substitution of non-renewable woody biomass by similar consumers. Use a value of 63.7 tCO₂/TJ
- $N_{y,i,j}$ Number of project devices of type i and batch j operating during year y .
- μ_y Adjustment to account for any continued use of pre-project devices during the year y applying equation 6

$$B_{y,savings,i,j} = B_{old,i,j} \times \left(1 - \frac{\eta_{old,i,j}}{\eta_{new,i,j}}\right)$$

Where:

⁷ For example, in some instances, full replacement of the pre-project device would require the implementation of more than one project device (e.g. one stove suitable for cooking and the other stove suitable for cooking/boiling water).

$\eta_{old,i,j}$ Efficiency of the old devices being replaced by project devices of type i and batch j.

$\eta_{new,i,j}$ Efficiency of the project device i and batch j

$B_{old,i,j}$ is determined as follows:

$$B_{old,i,j} = B_{old,HH} \div N_{d,HH}$$

Where:

$B_{old,HH}$ = Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices (tonnes/household/year)

$N_{d,HH}$ = Number of project devices per household (number)

$B_{old,i,j}$ is multiplied by a net to gross adjustment factor of 0.95 to account for leakages, in which case surveys are not required.

Jiko Matawi stove stove:

Substituting for values for Jiko Matawi stoves in use for 100% of the time over the monitoring period, the calculation results in the below.

	Value					Unit	Source/reference	Precision/confidence level
$N_{y,i,j}$	Year	Batch 1	Batch 2	Batch 3	Batch 4	n/a	Calculated $N_{y,i,j} = N * pop_stoves_y$	95/10 (biennial)
	2021	42,495	88,443	56,676	10,873			
	2022	42,495	88,443	56,676	10,873			
pop_stoves_y	Year	Batch 1	Batch 2	Batch 3	Batch 4	%	Monitoring surveys	95/10 (biennial)
	2021	96,77%	98,33%	97,50%	100%			
	2022	96,77%	98,33%	97,50%	100%			
μ_y	Year	Batch 1	Batch 2	Batch 3	Batch 4	fraction	Monitoring surveys see	95/10 (biennial)

	202 1	94,67 %	97,79 %	99,21 %	99,64 %		section 6.1 above	
	202 2	94,67 %	97,79 %	99,21 %	99,64 %			
fNRB,y	0.8183					fraction	Section E.1 of registered CPA1	N/A
NCVbiomass	0.0156					TJ/tonne	Section E.2 of registered CPA1	N/A
EFprojected_ fossilfuel	63.7					tCO ₂ /TJ	Methodology AMS-II.G version 10.0	N/A
By,savings,i,j	3.39					tonnes/y ear	ER calculations By,savings,i,j = Bold,i,j*(1 – ηold,i,j/ηnew ,i,j)*0.95	N/A
Bold,i,j	5.03					tonnes/y ear	Section E.1 of registered CPA1	N/A
Bold,HH	5.03					tonnes/H H/year	Section E.1 of registered CPA1	N/A
Nd,HH	1.0					Personne s/HH	Monitoring surveys see section 6.1 above	N/A
ηold,i,j	10					%	Section E.1 of registered CPA1	N/A

$\eta_{new,i,j}$	Year	Batch 1	Batch 2	Batch 3	Batch 4	%	Section E.2	95/10 (annually)
	2021	30.71 %	30.71 %	30.71 %	30.71 %			
	2022	30.71 %	30.71 %	30.71 %	30.71 %			
Baseline Emissions	Year	Batch 1	Batch 2	Batch 3	Batch 4	tCO ₂ /year	Section F.1, ER calculations $ER_{y,i,j} = B_{y,savings,i,j} \times N_{y,i,j} \times \mu_y \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected,fd}$	N/A
	2021	105,396	226,601	147,311	28,386			
	2022	25,988	55,874	36,323	6,999			
Project emissions (PE _y)	0					tCO ₂ /year	Section F.2, ER calculations	
Leakage emissions (LE _y)	0					tCO ₂ /year	Section E.1, section F.3 Adjustment factor (0.95) already applied at B _{y,savings,i,j} level	N/A
Emission reductions	Year	Batch 1	Batch 2	Batch 3	Batch 4	tCO ₂ /year	ER calculations $ER_y = \sum_i \sum_j ER_{y,i,j} - LE_y$	N/A
	2021	105,396	226,601	147,311	28,386			
	2022	25,988	55,874	36,323	6,999			

After taking account of the effective usage days of each individual stove after the date of sale⁸, effective emissions reductions are as follows.

6.3 Project Emissions

As per ER calculation, and the methodology, there are no project emissions taken account of.

6.4 Leakage

As per AMS II.G, leakage is already taken account of in the calculation of baseline emissions by multiplying By,savings,i,j by a net to gross adjustment factor of 0.95 to account for leakages. In such case, surveys to determine leakage ex-post are not required.

Further leakage emissions occur due to switching from baseline device using firewood to efficient project device using charcoal. These emissions are calculated by using a default value of 0.030 t CH₄/t charcoal in accordance with “AMS-III.BG.: Emission reduction through sustainable charcoal production and consumption”.

6.5 Net GHG Emission Reductions and Removals

The monitoring period for which GHG emission reductions were achieved spans from 01-January-2021 to 31-March-2022

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
01/01/2021-31/12/2021	507,694	0	0	507,694
01/01/2022-31/03/2022	125,184	0	0	125,184
Total	632,878	0	0	632,878

⁸ Cut-off date of this monitoring period is 31/03/2022.